

```
(dharani0612@kali)-[~]  
$ nano great.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x great.sh
```

```
(dharani0612@kali)-[~]  
$ ./great.sh
```

ENTER THREE NUMBERS

2 5 9

9 is greater

```
(dharani0612@kali)-[~]  
$ nano factorial.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x factorial.sh
```

```
(dharani0612@kali)-[~]  
$ bash factorial.sh
```

ENTER THE NUMBER:

5

FACTORIAL OF THE GIVEN NUMBER IS 120

```
(dharani0612@kali)-[~]  
$ nano oddsum.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x oddsum.sh
```

```
(dharani0612@kali)-[~]  
$ bash oddsum.sh
```

ENTER THE RANGE:

20

SUM = 100

```
(dharani0612@kali)-[~]  
$ nano fibonacci.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x fibonacci.sh
```

```
(dharani0612@kali)-[~]  
$ ./fibonacci.sh
```

ENTER THE LIMIT:

7

0

1

1

2

3

5

8

```
(dharani0612@kali)-[~]  
$ nano calculator.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x calculator.sh
```

```
(dharani0612@kali)-[~]  
$ ./calculator.sh
```

ENTER THE VALUE OF A:

3

ENTER THE VALUE OF B:

4

ENTER THE OPTION TO PERFORM

1. ADDITION

2. SUBTRACTION

3. MULTIPLICATION

4. DIVISION

1

Result = 7

```
(dharani0612@kali)-[~]  
$ nano largestdigit.sh
```

```
(dharani0612@kali)-[~]  
$ nano largestdigit.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x largestdigit.sh
```

```
(dharani0612@kali)-[~]  
$ ./largestdigit.sh
```

ENTER THE NUMBER

3456

THE LARGEST DIGIT OF THE NUMBER: 6

```
(dharani0612@kali)-[~]  
$ nano palindrome.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x palindrome.sh
```

```
(dharani0612@kali)-[~]  
$ ./palindrome.sh
```

ENTER THE STRING TO CHECK PALINDROME

malayalam

malayalam is a palindrome

```
(dharani0612@kali)-[~]  
$ nano reverse.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x reverse.sh
```

```
(dharani0612@kali)-[~]  
$ ./reverse.sh
```

ENTER THE NUMBER

456

REVERSE OF THE NUMBER IS 654

```
(dharani0612@kali)-[~]
$ nano parent.sh

(dharani0612@kali)-[~]
$ chmod +x parent.sh

(dharani0612@kali)-[~]
$ ./parent.sh
Parent Process ID : 27301
Child Process ID : 27301
Parent Process ID : 2226
Child Process Completed
```

```
(dharani0612@kali)-[~]
$ nano parentch.sh

(dharani0612@kali)-[~]
$ chmod +x parentch.sh

(dharani0612@kali)-[~]
$ bash parentch.sh
Child Process Running
Child Process Completed
Parent Resumes Execution
```

```
(dharani0612@kali)-[~]
$ touch sample.txt

(dharani0612@kali)-[~]
$ chmod +x parentch3.sh

(dharani0612@kali)-[~]
$ bash parentch3.sh
File Opened Successfully
File Closed Successfully
```

```
(dharani0612@kali)-[~]
$ bash ex4-1.sh
Enter Number of Processes
3
Enter Burst Time for P1
4
Enter Burst Time for P2
6
Enter Burst Time for P3
7

Process BT      WT      TAT
P1      4        0        4
P2      6        4       10
P3      7       10       17

Average Waiting Time = 4.67
Average Turnaround Time = 10.33
```

```
(dharani0612@kali)-[~]
$ bash ex4-2.sh
Enter Number of Processes
2
Enter Burst Time for P1
3
Enter Burst Time for P2
4

Process BT      WT      TAT
P1      3        0        3
P2      4        3        7

Average Waiting Time = 1.50
Average Turnaround Time = 5.00
```

```
(dharani0612@kali)-[~]
$ bash ex4-3.sh
Enter Number of Processes
2
Enter Burst Time for P1
3
Enter Priority for P1
4
Enter Burst Time for P2
3
Enter Priority for P2
5

Process Priority  BT      WT      TAT
P1      4        3        0        3
P2      5        3        3        6

Average Waiting Time = 1.50
Average Turnaround Time = 4.50
```

```
(dharani0612@kali)-[~]
$ chmod +x ex5.sh

(dharani0612@kali)-[~]
$ bash ex5.sh
Parent Received: Message from Child Process
```

```
(dharani0612@kali)-[~]
$ nano ex6.sh

(dharani0612@kali)-[~]
$ chmod +x ex6.sh

(dharani0612@kali)-[~]
$ bash ex6.sh
Entering Critical Section
Leaving Critical Section
```



```
(dharani0612@kali)-[~]
$ bash ex4-4.sh
Enter Number of Processes
2
Enter Burst Time for P1
4
Enter Burst Time for P2
5
Enter Time Quantum
2

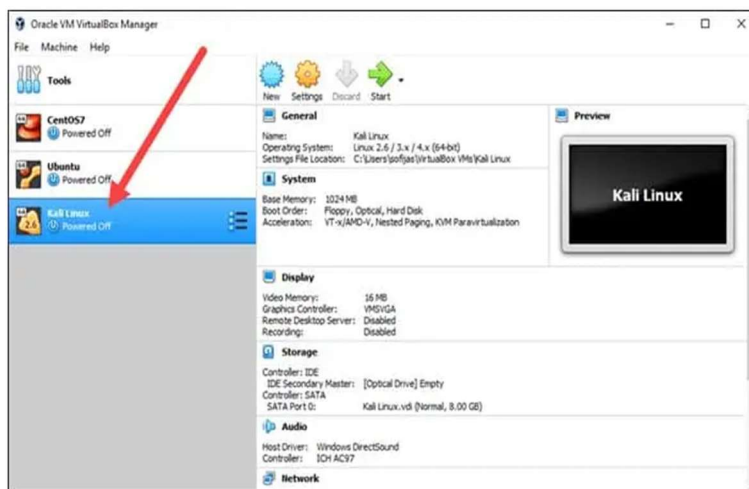
Process BT      WT      TAT
P1      4        2        6
P2      5        4        9

Average Waiting Time = 3.00
Average Turnaround Time = 7.50
```

```
(dharani0612@kali)-[~]
$ nano ex7.sh

(dharani0612@kali)-[~]
$ chmod +x ex7.sh

(dharani0612@kali)-[~]
$ bash ex7.sh
Enter Number of Processes:
3
Enter Safe Sequence (space separated):
2 3 4
Safe Sequence is:
P2 P3 P4
System is in Safe State
```



```
(dharani0612@kali)-[~]
$ nano ex8.sh

(dharani0612@kali)-[~]
$ chmod +x ex8.sh

(dharani0612@kali)-[~]
$ bash ex8.sh
Enter number of processes:
3
Enter deadlocked process numbers (if any):
2
Deadlocked Processes:
2
```



```

(dharani0612@kali)-[~]
$ nano ex9.sh

(dharani0612@kali)-[~]
$ chmod +x ex9.sh

(dharani0612@kali)-[~]
$ bash ex9.sh
Thread 1 : 1
Thread 2 : 1
Thread 1 : 2
Thread 2 : 2
Thread 1 : 3
Thread 2 : 3
Thread 1 : 4
Thread 2 : 4
Thread 2 : 5
Thread 1 : 5
All Threads Completed

```

```

(dharani0612@kali)-[~]
$ nano ex10.sh

(dharani0612@kali)-[~]
$ chmod +x ex10.sh

(dharani0612@kali)-[~]
$ bash ex10.sh
Enter Page Size:
4
Enter Number of Pages:
8
Enter Frame Number for Page 0:
2
Enter Frame Number for Page 1:
3
Enter Frame Number for Page 2:
4
Enter Frame Number for Page 3:
5
Enter Frame Number for Page 4:
6
Enter Frame Number for Page 5:
7
Enter Frame Number for Page 6:
8
Enter Frame Number for Page 7:
9
Enter Logical Address:
7
Page Number : 1
Offset : 3
Frame Number : 3
Physical Address : 15

```

Message received from child: Hello from Child Process

...Program finished with exit code 0
Press ENTER to exit console.

```

(dharani0612@kali)-[~]
$ nano ex11-1.sh

(dharani0612@kali)-[~]
$ chmod +x ex11-1.sh

(dharani0612@kali)-[~]
$ bash ex11-1.sh
First Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using First Fit

```

```

(dharani0612@kali)-[~]
$ nano ex11-2.sh

(dharani0612@kali)-[~]
$ chmod +x ex11-2.sh

(dharani0612@kali)-[~]
$ bash ex11-2.sh
Best Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using Best Fit

```



```
(dharani0612@kali)-[~]  
$ nano ex11-3.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x ex11-3.sh
```

```
(dharani0612@kali)-[~]  
$ bash ex11-3.sh
```

Worst Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using Worst Fit

```
(dharani0612@kali)-[~]  
$ nano ex12.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x ex12.sh
```

```
(dharani0612@kali)-[~]  
$ bash ex12.sh
```

FIFO Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
Frames: 3
FIFO Algorithm Executed

```
(dharani0612@kali)-[~]  
$ nano ex12-2.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x ex12-2.sh
```

```
(dharani0612@kali)-[~]  
$ bash ex12-2.sh
```

LRU Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
LRU Algorithm Executed

```
(dharani0612@kali)-[~]  
$ nano ex12-3.sh
```

```
(dharani0612@kali)-[~]  
$ chmod +x ex12-3.sh
```

```
(dharani0612@kali)-[~]  
$ bash ex12-3.sh
```

Optimal Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
Optimal Algorithm Executed

```
Enter Number of Pages: 3  
Enter Reference String:  
3 4 5  
Enter Number of Frames: 3  
Total Page Faults = 3
```

```
Enter Number of Pages: 3  
Enter Reference String:  
3 3 3  
Enter Number of Frames: 3  
Total Page Faults = 1
```

```
Enter Number of Pages: 4  
Enter Reference String:  
3 4 3 3  
Enter Number of Frames: 2  
Total Page Faults = 2
```

```
Enter Number of Blocks: 4  
Enter Number of Processes: 4  
Enter Block Sizes:  
3 2 3 2  
Enter Process Sizes:  
4 3 4 3
```

Process No	Process Size	Block No
1	4	Not Allocated
2	3	1
3	4	Not Allocated
4	3	3

```
Enter Number of Blocks: 4
Enter Number of Processes: 4
Enter Block Sizes:
3 4 5 3
Enter Process Sizes:
3 4 5 6
```

Process No	Process Size	Block No
1	3	1
2	4	2
3	5	3
4	6	Not Allocated

```
Enter Number of Blocks: 4
Enter Number of Processes: 4
Enter Block Sizes:
3 3 2 2
Enter Process Sizes:
4 3 4 3
```

Process No	Process Size	Block No
1	4	Not Allocated
2	3	1
3	4	Not Allocated
4	3	2

```
Enter Page Size: 4
Enter Number of Pages: 5
Enter Frame Numbers for Each Page:
Page 0 -> Frame: 3
Page 1 -> Frame: 4
Page 2 -> Frame: 5
Page 3 -> Frame: 3
Page 4 -> Frame: 4
Enter Logical Address: 3
```

```
Page Number      : 0
Offset           : 3
Frame Number      : 3
Physical Address  : 15
```

```
Thread Executing : 1
Thread Executing : 1
Thread Executing : 2
Thread Executing : 2
Thread Executing : 3
Thread Executing : 3
Thread Executing : 4
Thread Executing : 4
Thread Executing : 5
Thread Executing : 5
All Threads Completed
```

```
File Opened Successfully
File Closed Successfully
```

```
Child Process Running
Child Process Completed
Parent Resumes Execution
```

```
Parent PID : 3218
```

```
PARENT PROCESS
```

```
Parent PID : 3218
```

```
Parent's Parent PID : 3217
```

```

Enter Number of Processes: 3
Enter Number of Resource Types: 2

Enter Allocation Matrix:
0 1
1 0
1 1

Enter Request Matrix:
1 0
0 1
0 0

Enter Available Resources:
0 0

Deadlocked Processes:
No Deadlock Detected

```

```

Enter Number of Processes: 5
Enter Number of Resources: 3

```

```

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2

```

```

Enter Maximum Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3

```

```

Enter Available Resources:
3 3 2

```

```

System is in Safe State
Safe Sequence: P1 P3 P4 P0 P2

```

```

Parent Process Leaving Critical Section
Child Process Entering Critical Section
Child Process Leaving Critical Section

```

```

...Program finished with exit code 0
Press ENTER to exit console.

```

```

Enter Number of Processes: 3
Enter Burst Time for P1: 5
Enter Burst Time for P2: 3
Enter Burst Time for P3: 4
Enter Time Quantum: 2

```

Process	BT	WT	TAT
P1	5	7	12
P2	3	6	9
P3	4	7	11

```

Average Waiting Time      = 6.67
Average Turnaround Time  = 10.67

```

```

...Program finished with exit code 0
Press ENTER to exit console.

```


Enter Number of Processes: 3

Enter Burst Time for P1: 6

Enter Priority for P1: 2

Enter Burst Time for P2: 8

Enter Priority for P2: 1

Enter Burst Time for P3: 7

Enter Priority for P3: 3

Process	Priority	BT	WT	TAT
P2	1	8	0	8
P1	2	6	8	14
P3	3	7	14	21

Average Waiting Time = 7.33

Average Turnaround Time = 14.33

...Program finished with exit code 0

Press ENTER to exit console.

Enter Number of Processes: 4

Enter Burst Time for P1: 6

Enter Burst Time for P2: 2

Enter Burst Time for P3: 8

Enter Burst Time for P4: 3

Process	BT	WT	TAT
P2	2	0	2
P4	3	2	5
P1	6	5	11
P3	8	11	19

Average Waiting Time = 4.50

Average Turnaround Time = 9.25

...Program finished with exit code 0

Press ENTER to exit console.

Enter Number of Processes: 3

Enter Burst Time for P1: 24

Enter Burst Time for P2: 3

Enter Burst Time for P3: 3

Process	BT	WT	TAT
P1	24	0	24
P2	3	24	27
P3	3	27	30

Average Waiting Time = 17.00

Average Turnaround Time = 27.00

...Program finished with exit code 0

Press ENTER to exit console.


```
Applications Places Terminal Jul 18 12:21 PM
kali@kali: ~/OS/exp13

(kali@kali)-[~/OS/exp13]
$ gcc fileorganization.c -o fileorg

(kali@kali)-[~/OS/exp13]
$ ./fileorg
----- SEQUENTIAL FILE ORGANIZATION -----
Enter Register Number: 101
Enter Name: Arun
Record Details
Register Number : 101
Name : Arun

(kali@kali)-[~/OS/exp13]
$ ./direct
----- DIRECT FILE ORGANIZATION -----
Enter Register Number: 101
Enter Name: Arun
Record Found
Reg No : 101
Name : Arun

(kali@kali)-[~/OS/exp13]
$ ./indexed
----- INDEXED FILE ORGANIZATION -----
Enter number of records: 3
Enter details (RegNo Name):
101 Arun
102 Kumar
103 Ravi
Enter Register Number to Search: 102
Record Found
Reg No : 102
Name : Kumar

(kali@kali)-[~/OS/exp13]
$
```

EXPERIMENT 13 – FILE ORGANIZATION TECHNIQUES

```
Applications Places Terminal Jul 18 12:26 PM
kali@kali: ~/OS/exp14

(kali@kali)-[~/OS/exp14]
$ gcc allocation.c -o alloc

(kali@kali)-[~/OS/exp14]
$ ./alloc
----- SEQUENTIAL ALLOCATION -----
Enter Starting Block: 10
Enter File Length (Number of Blocks): 5
Allocated Blocks:
10 11 12 13 14

(kali@kali)-[~/OS/exp14]
$ ./indexed
----- INDEXED ALLOCATION -----
Enter Index Block: 20
Enter Number of Blocks: 4
Enter Block Numbers:
5
8
12
15
Index Block : 20
Allocated Blocks : 5 8 12 15

(kali@kali)-[~/OS/exp14]
$ ./linked
----- LINKED ALLOCATION -----
Enter Number of Blocks: 4
Enter Block Numbers:
10
18
25
31
Linked Allocation:
10 --> 18 --> 25 --> 31 --> NULL

(kali@kali)-[~/OS/exp14]
$
```

EXPERIMENT 14 – FILE ALLOCATION STRATEGIES

```
Applications Places Terminal Jul 18 12:31 PM
kali@kali: ~/OS/exp15

(kali@kali)-[~/OS/exp15]
$ gcc fcfs.c -o fcfs

(kali@kali)-[~/OS/exp15]
$ ./fcfs
Enter Number of Requests: 8
Enter Request Queue:
98 183 37 122 14 124 65 67
Enter Initial Head Position: 53
Total Head Movement = 640

(kali@kali)-[~/OS/exp15]
$ gcc sstf.c -o sstf

(kali@kali)-[~/OS/exp15]
$ ./sstf
Enter Number of Requests: 8
Enter Request Queue:
98 183 37 122 14 124 65 67
Enter Initial Head Position: 53
Total Head Movement = 236

(kali@kali)-[~/OS/exp15]
$ gcc scan.c -o scan

(kali@kali)-[~/OS/exp15]
$ ./scan
SCAN Disk Scheduling
Initial Head Position: 53
Head moves towards higher cylinders,
then reverses direction.

(kali@kali)-[~/OS/exp15]
$ gcc cscan.c -o cscan

(kali@kali)-[~/OS/exp15]
$ ./cscan
C-SCAN Disk Scheduling
Initial Head Position: 53
Head moves in one direction only.
After reaching the end,
it returns to the beginning.

(kali@kali)-[~/OS/exp15]
$
```

EXPERIMENT 15 – DISK SCHEDULING ALGORITHMS